

Simulating Chromosomes with OpenMiChroM



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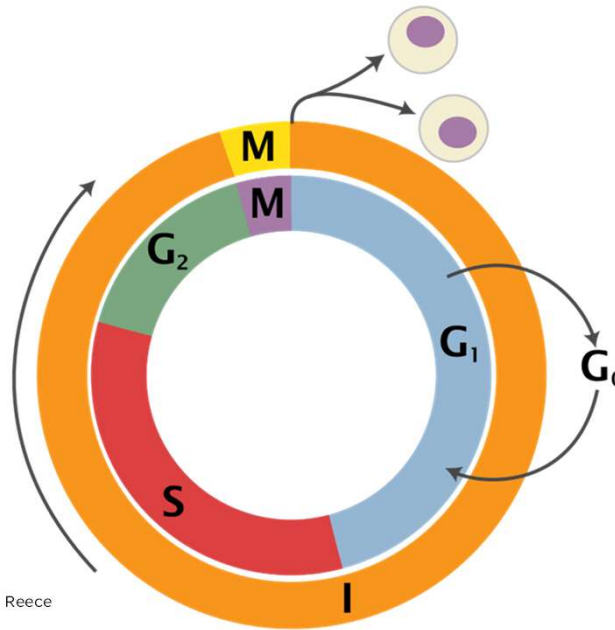
Cell Cycle – DT40

I

Interphase ~ 95% living time

M

Mitotic phase - Nuclear division



By Ali Zifan - from:Campbell Biology (10th Edition) by: Jane B. Reece
& Steven A. Wasserman.and Nature.com., CC BY-SA 4.0

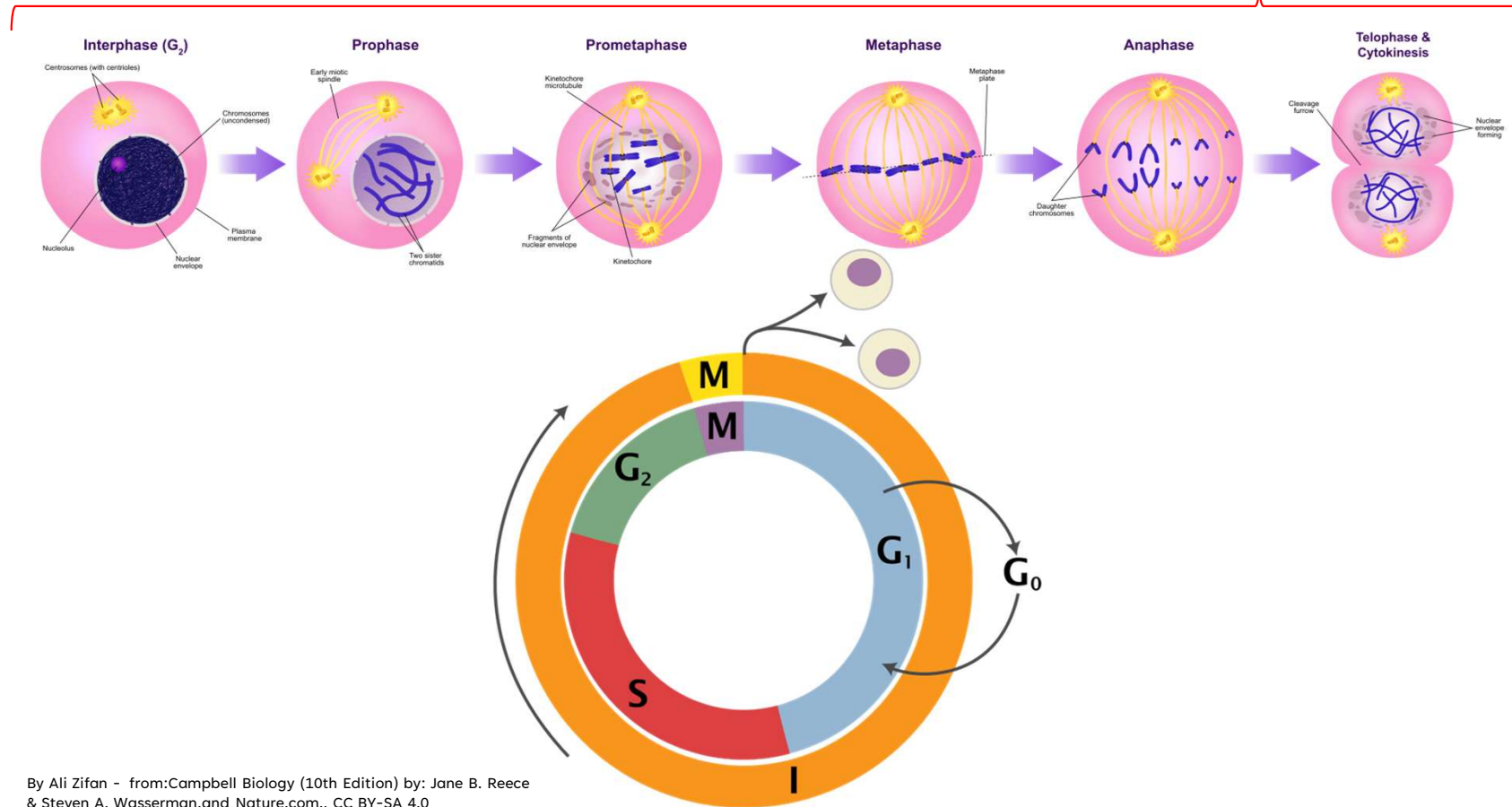
Cell Cycle – DT40

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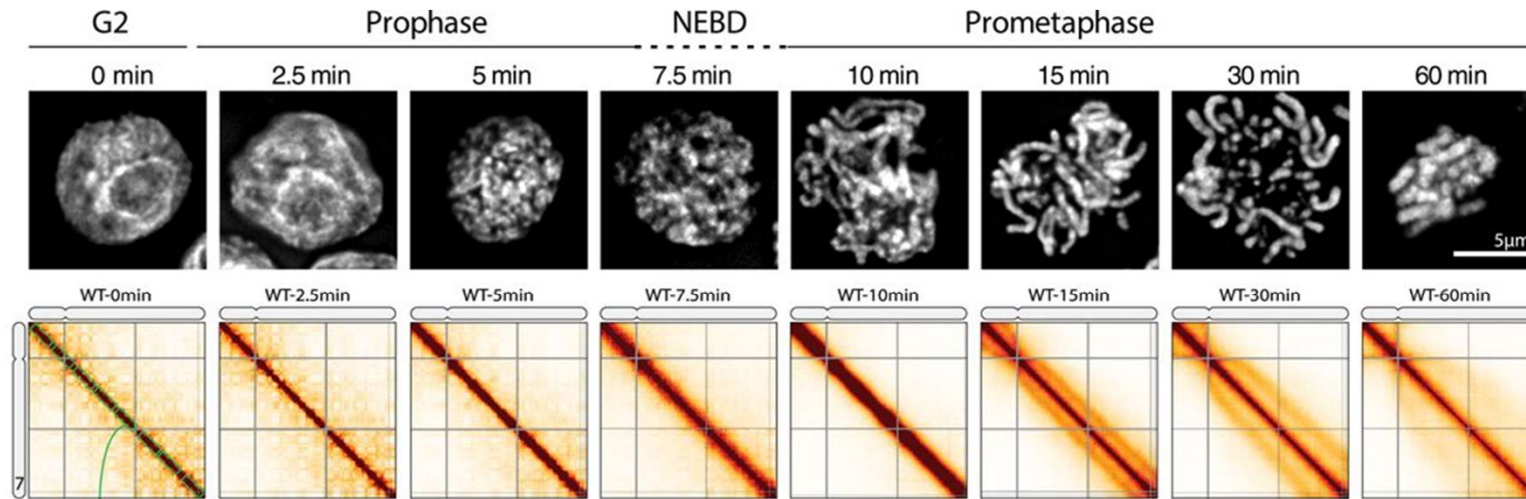
M

Mitotic phase - Nuclear division



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Cell Cycle – DT40

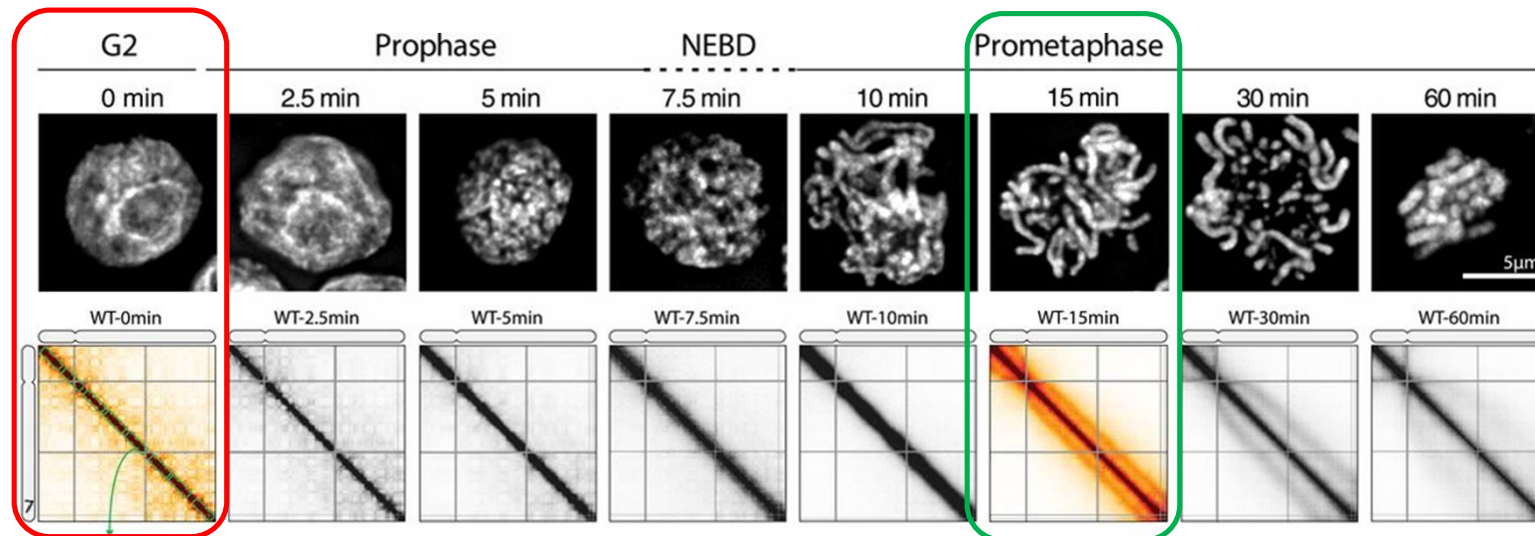


Gibcus J. H., Dekker J. et al. A pathway for mitotic chromosome formation ,Science (2018)



Nuclear Envelope Breakdown

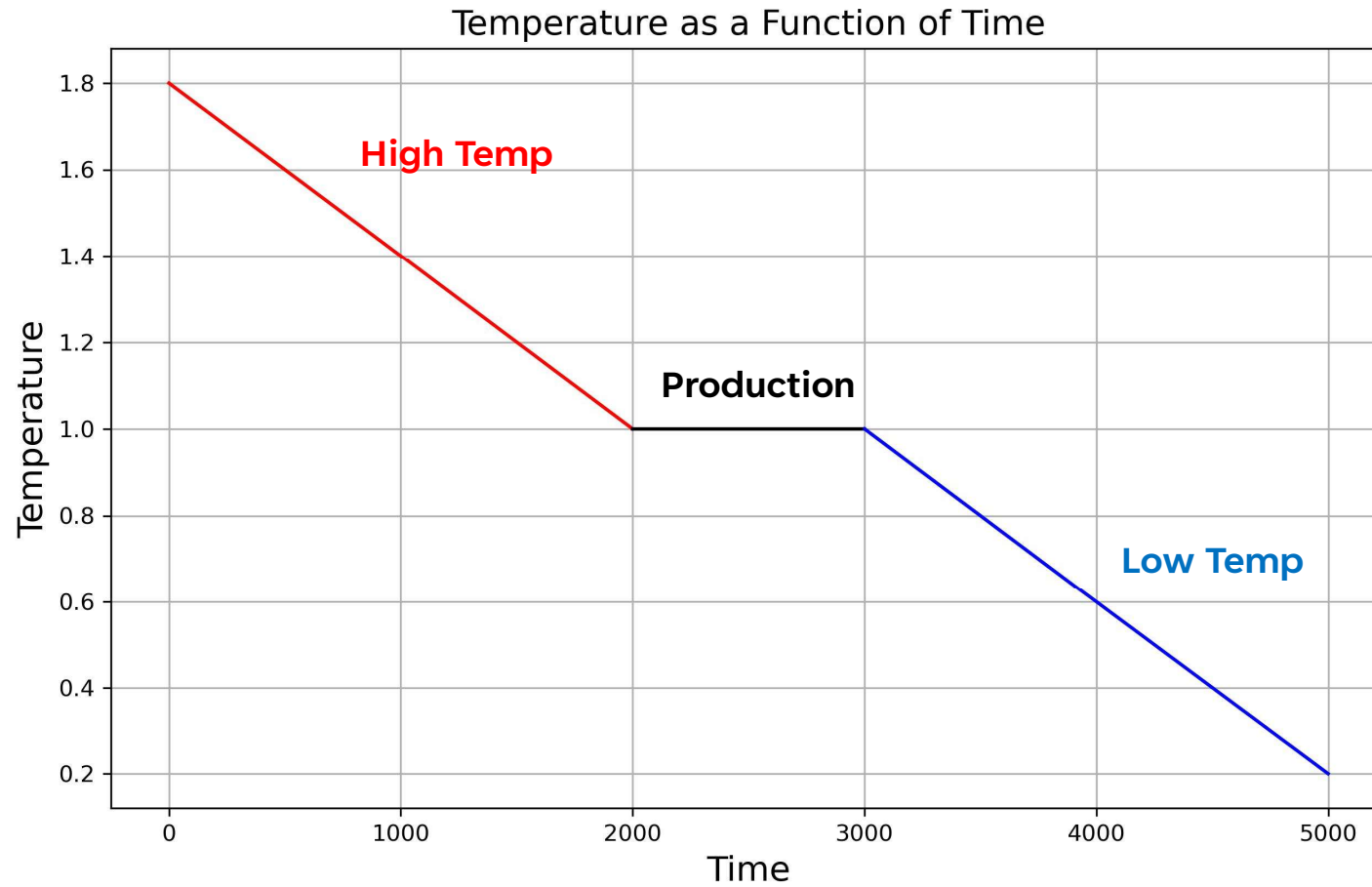
Cell Cycle – DT40



Gibcus J. H., Dekker J. et al. A pathway for mitotic chromosome formation ,Science (2018)

- Chromosome 7 @ 50kb resolution
- Interphase
- Prometaphase
- Quenching Simulations

Quenching Simulations



VSCode

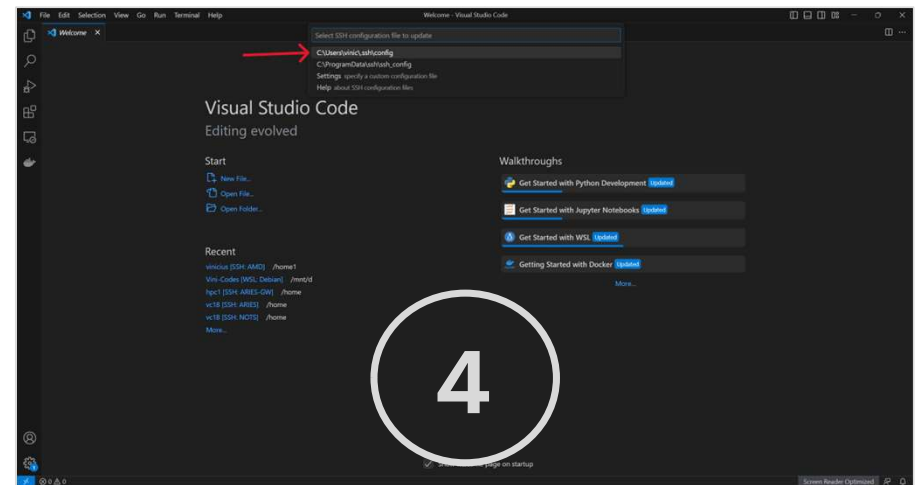
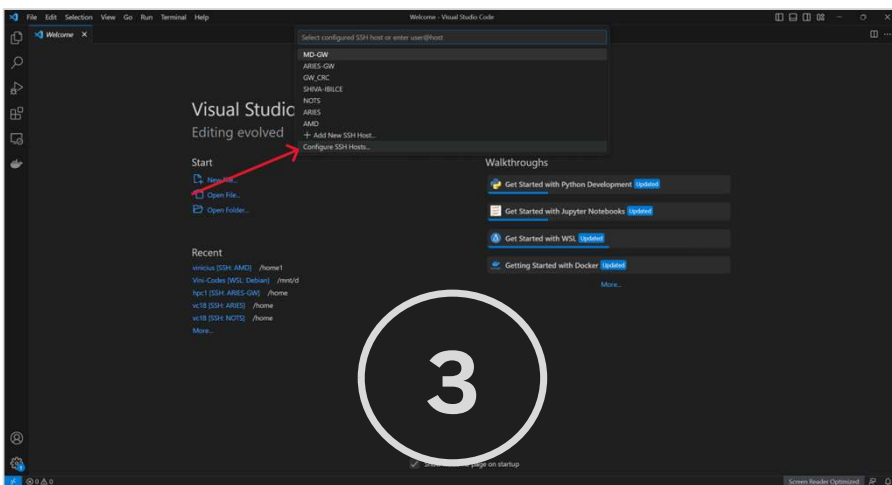
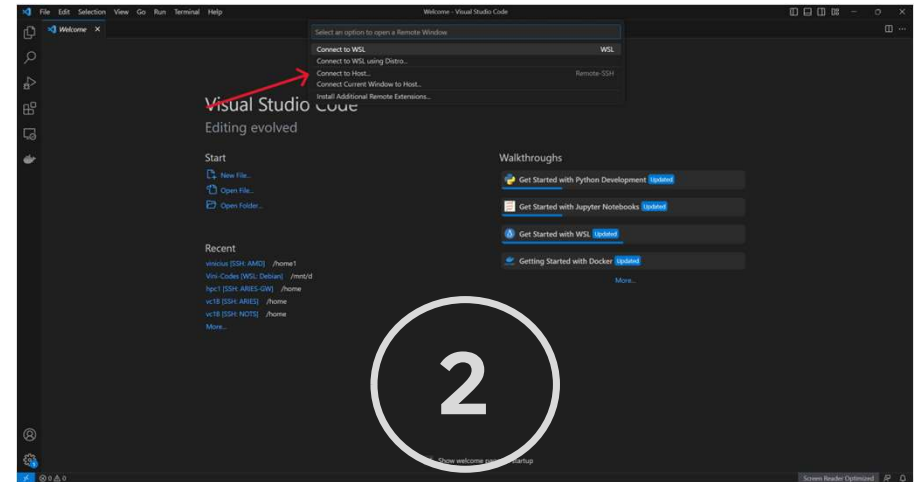
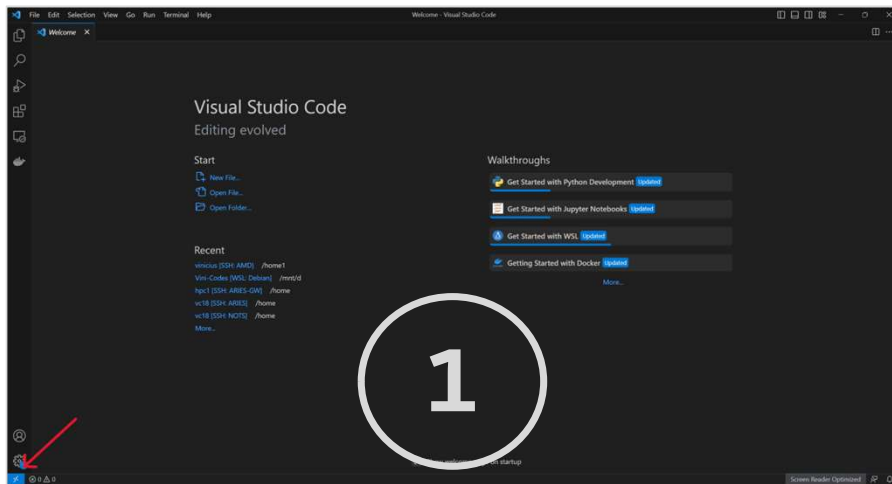
Install Extensions:

- SSH remote
- Jupyter / Python
- Open Files and Terminal

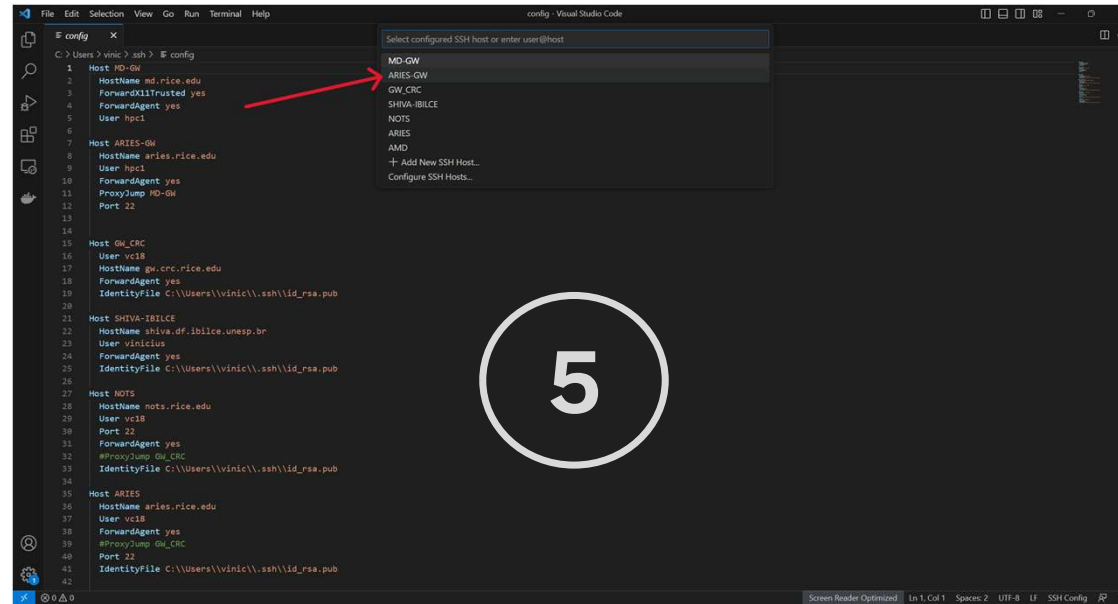
```
Host MD-GW
  HostName md.rice.edu
  ForwardX11Trusted yes
  ForwardAgent yes
  User hpc1

Host ARIES-GW
  HostName aries.rice.edu
  User hpc1
  ForwardAgent yes
  ProxyJump MD-GW
  Port 22
```

VSCode



VSCode



```
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```

Download Files

- Download from https://smog-server.org/workshops/CTBP_2023
- Unzip the folder in your computer - Desktop
- Send folder/files to the Aries cluster using VSCode – Drag and Drop

```
cd Workshop_OpenMiChroM_CTBP_June_2023/  
[hpc1@ln-mgt3 Workshop_OpenMiChroM_CTBP_June_2023]$ ls  
bin  inputs  
[hpc1@ln-mgt3 Workshop_OpenMiChroM_CTBP_June_2023]$ cd bin/  
[hpc1@ln-mgt3 bin]$ ls  
analysis.ipynb quenching_interphase.py quenching_mitotic.py  
[hpc1@ln-mgt3 bin]$ code quenching_interphase.py
```

Code Overview

```
# Import libraries
from OpenMiChroM.ChromDynamics import MiChroM
from OpenMiChroM.CndbTools import cndbTools
import numpy as np
import scipy as sc
import pandas as pd

# Call MiChroM module
sim = MiChroM(temperature=1.0, time_step=0.01)

# Setup platform
# HIP or OpenCL - AMD GPUs
# CUDA or OpenCL - NVidia
sim.setup(platform="HIP")
```

Code Overview

```
# Path to save the simulation outputs
data_folder = "/home/hpc1/Vini/"
sim.saveFolder(data_folder + 'output_quenching_interphase')

# Path to the chromatin sequence file
seq_File = "/home/hpc1/Vini/inputs/DT40_chr7_50kb.seq"

# Create and load the initial structure - Line
interphase_initial_structure = sim.createLine(ChromSeq=seq_File)
sim.loadStructure(interphase_initial_structure, center=True)
```

Code Overview

```
# Add Homopolymer Potentials
sim.addFENEbonds(kfb=30.0)
sim.addAngles(ka=2.0)
sim.addRepulsiveSoftCore(Ecut=4.0)

# Path to the types interaction matrix
types_File = "/home/hpc1/Vini/inputs/types_DT40_chr7_50kb_interphase.csv"
# Add Types interaction
sim.addCustomTypes(mu=1.51, rc=2.12, TypesTable=types_File)

# Path to the Ideal Chromosome potential
IC_File = "/home/hpc1/Vini/inputs/IC_DT40_chr7_50kb_interphase.dat"
# Add Ideal Chromosome interaction
sim.addCustomIC(mu=1.51, rc=2.12, dinit=3, dend=303, IClist= IC_File)
```

Code Overview

```
# Save Structures - Gro and CNDB
sim.saveStructure(mode = 'gro')
sim.initStorage(filename = "traj_interphase")

# Temperature Parameters
Tstart = 1.8
Tmid = 1.0
Tend = 0.2
mid_steps = 1000
total_steps = 5000
first_step = 2000 # Specify the step at which you want to reach T=1.0

# Calculate the steps before, during, and after the mid_steps
start_steps = first_step
end_steps = total_steps - mid_steps - start_steps

# Create the temperature array
T = np.concatenate([
    np.linspace(Tstart, Tmid, start_steps, endpoint=False),
    np.full(mid_steps, Tmid),
    np.linspace(Tmid, Tend, end_steps)
])
```

Code Overview

```
# Simulation steps
block = 2*10**3

# List of temperatures to save structure at
save_temps = [1.5, 1.0004, 0.5]

for i in T:
    print("Temperature:", i)
    sim.integrator.setTemperature(i/0.008314) # Set the Temperature value
    sim.runSimBlock(block, increment=True)
    sim.saveStructure() # Save CNDB
    # Check if the current temperature is in the save_temps list
    if i in save_temps:
        sim.saveStructure(mode = 'gro') # Save gro file

sim.saveStructure(mode = 'gro') # Save gro file
sim.storage[0].close()
```

Run

```
$ ./bin/slurmJob.sh
#or
$ srun --reservation=workshop -p debug --gres=gpu:1 --ntasks=1 --oversubscribe --pty bash

#Running Simulation - Interphase
$ singularity exec /home/hpc1/ctbp.sif python3 bin/quenching_interphase.py

# Running Simulation - Mitotic
$ singularity exec /home/hpc1/ctbp.sif python3 bin/quenching_mitotic.py
```

Simulation Info Output

```
*  
Temperature: 1.8  
Number of exceptions: 738  
adding force FENEBond 0  
adding force AngleForce 1  
Add exclusions for RepulsiveSoftCore force  
adding force RepulsiveSoftCore 2  
Add exclusions for CustomTypes force  
adding force CustomTypes 3  
Add exclusions for CustomIC force  
adding force CustomIC 4  
Positions... loaded!  
potential energy is 22.377844
```

Simulation Info Output

Block number	Position – bead 1	Displacement	Time	Kinetic Energy	Potential Energy	Radius of Gyration	Steps per second
--------------	-------------------	--------------	------	----------------	------------------	--------------------	------------------

```
Temperature: 0.7750875437718859
bl=3563 pos[1]=[-7.6 3.2 14.1] dr=1.84 t=71260.0ps kin=1.17 pot=20.04 Rg=5.608 SPS=10643
Temperature: 0.7746873436718359
bl=3564 pos[1]=[-7.3 2.4 11.1] dr=1.75 t=71280.0ps kin=1.15 pot=19.99 Rg=5.504 SPS=11565
Temperature: 0.7742871435717859
bl=3565 pos[1]=[-10.8 4.7 15.8] dr=1.85 t=71300.0ps kin=1.12 pot=20.01 Rg=5.573 SPS=11007
Temperature: 0.7738869434717359
bl=3566 pos[1]=[-10.6 2.2 12.5] dr=1.87 t=71320.0ps kin=1.18 pot=19.98 Rg=5.462 SPS=11738
Temperature: 0.7734867433716859
bl=3567 pos[1]=[-10.8 1.5 13.0] dr=1.88 t=71340.0ps kin=1.18 pot=20.01 Rg=5.584 SPS=11798
```

Simulation Info Output

Statistics for the simulation Chromosome, number of particles: 739, number of chains: 1

Statistics for particle position

mean position is: [-17.86048866 -5.87324508 10.98828839] Rg = 4.7121344

median bond size is 0.962432837214557

three shortest/longest (<10)/ bonds are [0.91293946 0.91418144 0.91871809] [1.01948842 1.0233238 1.02487089]

95 percentile of distance to center is: 6.306322735430825

density of closest 95% monomers is: 0.6682682939084371

density of the core monomers is: 0.890598692481511

min/median/mean/max coordinates are:

x: -23.40, -17.88, -17.86, -11.57

y: -11.88, -5.85, -5.87, 0.09

z: 4.04, 10.99, 10.99, 17.39

Statistics for velocities:

mean kinetic energy is: 0.47556418454204363 should be: 1.5

fastest particles are (in kT): [2.06612597 2.06928829 2.13152468 2.47386422 2.91575784]

Statistics for the system:

Forces are: ['FENEBond', 'AngleForce', 'RepulsiveSoftCore', 'CustomTypes', 'CustomIC']

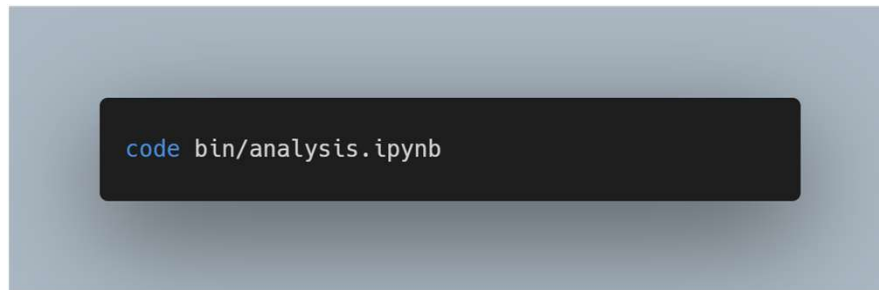
Number of exceptions: 738

Potential Energy Ep = 18.796201094238942

Analysis - CNDBTools

VSCode - Jupyter Notebook

- + button to open a new terminal window



- Environment myenv
- Shift + Enter -> run a cell

Visualize - VMD

- Navigate to your output folder in VScode – Aries
- Right-click and download the .gro file for your Desktop
- Open the structural file using VMD
 - Graphics -> Representation -> Drawing Method -> Tube

```
Display: No Depth Cueing  
Camera: Ortographic  
Color: CET_R2  
Material: A0Chalky  
Drawing Method: Tube  
Coloring Method: Index  
Radius: 3.5
```

Rendering High-Resolution image on Linux:

```
/usr/local/lib/vmd/tachyon_LINUXAMD64 -aasamples 12 image.dat -res  
2000 2000 -fullshade -auto_skylight 1.3 -format TARGA -o image.tga
```



Analysis - CNDBTools

Create Python environment – VSCode Jupyter Notebook

- `pip3 install virtualenv --user`
- `python3 -m virtualenv myenv`
- `source myenv/bin/activate`
- `pip3 install OpenMiChroM matplotlib numpy seaborn ipykernel`
- `python3 -m ipykernel install --user --name=myenv`
- `code analysis.ipynb`

- Ctrl + Shift + P – Python: Select Interpreter
- Enter interpreter path – Find : `/home/hpc1/myenv/bin/python3`

References

- <https://open-michrom.readthedocs.io> – Tutorials
- <https://github.com/junioreif/OpenMiChroM> - GitHub
- **Open-MiChroM** - [1] Oliveira Jr, A. B., Contessoto, V. G., Mello, M. F., & Onuchic, J. N. (2020). A scalable computational approach for simulating complexes of multiple chromosomes. *Journal of Molecular Biology*, 433(6), 166700.
- **Optimization** - [2] Oliveira Jr, A. B., et al. “Chromosome Modeling on Downsampled Hi-C Maps Enhances the Compartmentalization Signal.” *The Journal of Physical Chemistry B*, vol. 125, no. 31, Aug. 2021, pp. 8757–67.
- **CNDB file format** - [3] Contessoto, Vinícius G, et al. “The Nucleome Data Bank: Web-Based Resources to Simulate and Analyze the Three-Dimensional Genome.” *Nucleic Acids Research*, vol. 49, no. D1, Jan. 2021, pp. D172–82.